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A novel two-stage method for well-diversified portfolio construction based on stock return prediction using machine learning☆

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ABSTRACT

The performance of portfolio model can be improved by introducing stock prediction based on machine learning methods. However, the prediction error is inevitable, which may bring losses to investors. To limit the losses, a common strategy is diversification, which involves buying low-correlation stocks and spreading the funds across different assets. In this paper, a diversified portfolio selection method based on stock prediction is proposed, which includes two stages. To be specific, the purpose of the first stage is to select diversified stocks with high predicted returns, where the returns are predicted by machine learning methods, i.e. random forest (RF), support vector regression (SVR), long short-term memory networks (LSTM), extreme learning machine (ELM) and back propagation neural network (BPNN), and the diversification level is measured by Pearson correlation coefficient. In the second stage, the predictive results are incorporated into a modified mean–variance (MMV) model to determine the proportion of each asset. Using China Securities 100 Index component stocks as study sample, the empirical results demonstrate that the RF+MMV model achieves better results than similar counterparts and market index in terms of return and return–risk metrics.

1. Introduction

Investment in the stock market helps to generate more returns than other financial instruments but has the threat of market risk that may lead to a high loss. Since investors naturally like returns and hate risks, they hope to spread funds reasonably to high-quality assets, so as to achieve the purpose of maximizing returns and minimizing risks (Guerard, Markowitz, & Xu, 2015). As we all know, stock quality is significantly contingent upon its future performance. Thus, accurate return prediction can help investors achieve high returns and hedge risks.

As the stock market is a nonlinear, evolutionary, dynamic and complex system in nature, stock prediction is a challenging problem in time series prediction. In the past, many scholars used time series models for stock prediction (Fatima, 2008; Jiang, Ding, Xu, & Tong, 2019; Mohammadi, 2017; Wang, Wang, Zhang, & Guo, 2012), such as linear regression, autoregressive integrated moving average (ARIMA), generalized autoregressive conditional heteroscedasticity (GARCH). Recently, many studies have demonstrated that machine learning models, such as support vector regression (SVR) (Huang, 2012; Ismail, Noorani, Ismail, Razak, & Alias, 2020;

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Mishra, Padhy, Mishra, & Misra, 2021; Pan, Xiao, Wang, & Yang, 2017), tree-based models (Basak, Kar, Saha, Khaidem, & Dey, 2019; Jiang, Liu, Zhang, & Liu, 2020; Nobre & Neves, 2019; Thakur & Kumar, 2018), deep learning (Fischer & Krauss, 2018a; Gocken, Ozcalici, Boru, & Dosdogru, 2019; Gunduz, Yaslan, & Cataltepe, 2017; U., Lu, Kim, Ryu, & Pak, 2020), artificial neural networks (ANNs) (Chen, Jiang, Zhang, & Jia, 2021; Hu, Tang, Zhang, & Wang, 2018; Pal & Kar, 2017; Qiu, Song, & Akagi, 2016) and so on, have a greater ability to treat problems with complex, non-stationary and nonlinear characteristics than traditional time series models. In this paper, five representative machine learning models, i,e, random forest (RF), support vector regression (SVR), long short-term memory networks (LSTM), extreme learning machine (ELM), as well as back propagation neural network (BPNN), are used for stock return prediction.

However, the prediction error is inevitable, which may bring losses to investors. To limit losses, a common strategy is diversification, which involves buying low-correlation stocks and spreading the fund across different assets. Therefore, many scholars constructed diversified portfolios to reduce unsystematic risks. For example, Nanda, Mahanty, and Tiwari (2010) developed a model for diversified stock selection, in which K-means, Fuzzy C-means and self organizing maps were used for clustering stock data. Baralis, Cagliero, and Garza (2017) presented an itemset-based approach to plan diversified stock portfolios in a buy-and-hold scenario by exploiting stock combinations, i.e., itemsets, mined from historical stock data. Wang, Li, and Guan (2019) analyzed the relationship between stock price correlation and product similarity, and concluded that product similarity can be used as a supplement for stock price correlation, which provide a reference for investors to make portfolio strategies. Schmidt (2019) proposed the diversified mean–variance portfolio (DMVT) theory in which a diversity booster term was incorporated explicitly into the objective function. Tayalı (2020) proposed a novel backtesting approach that integrates the clustering techniques into the MV portfolio optimization model to create diversified portfolios. This paper thus aims to select stocks that maximize returns while minimizing risks caused by insufficient diversification.

After the pre-selection process, the capital allocation proportion for each selected stocks needs to be determined. The seminal mean–variance (MV) model proposed by Markowitz (1952) serves as the basis for the development of the modern portfolio theory. Quantifying investment return as the mean of returns, and investment risk as the variance from the mean, Markowitz formulated MV model mathematically in two ways: minimizing variance for a given expected value, or maximizing expected value for a given variance. However, MV model has some limitations in practical applications, such as computational complexity and strict hypotheses for larger-scale assets (Elton & Gruber, 1973; Grootveld & Hallerbach, 1999). Therefore, some researchers further improved and promoted the MV model (Akbay, Kalayci, & Polat, 2020; Hakansson, 1971; Jin, Chen, & Yuan, 2018; Konno & Yamazaki, 1991; Zhu & Zhang, 2021).

Recently, taking the advantages of machine learning models in stock forecasting, many researchers incorporate these models into portfolio optimization models to improve portfolio performance. Mishra, Panda, and Majhi (2016) advanced a prediction-based mean–variance (PBMV) model, in which heuristic functional link artificial neural network (HFLANN) was applied to forecast future returns, and the prediction results were used to the constrained portfolio optimization model. Almahdi and Yang (2017) presented a recurrent reinforcement learning method with expected maximum drawdown to simultaneously generate both buy/sell signals and optimal portfolio selection. Paiva, Cardoso, Hanaoka, and Duarte (2019) developed an investment decision model using support vector machine (SVM) method for asset classification and MV model for portfolio selection. Mishra and Padhy (2019) proposed an efficient portfolio construction method based on stock price prediction by the SVR model and stock selection by the MV model. Yu, Chiou, Lee, and Lin (2020) integrated the ARIMA prediction model with different portfolio models, including the MV, downside risk (DSR), conditional value-at-risk (CVaR), value-at-risk (VaR), mean-absolute deviation (MAD) and Omega models. Yun, Lee, Kang, and Seok (2020) proposed a two-stage deep learning framework called Grouped-ETFs Model (GEM) and a joint cost function to train the deep learning model for portfolio construction. Wang, Li, Zhang, and Liu (2020) presented a novel method for portfolio formation where LSTM method was used to predict and select assets with a higher daily return of gain, and the MV method was applied to form optimal portfolios. Chen, Zhong, and Chen (2020) solved the mean–variance–skewness–kurtosis (M–V–S–K) model by developing a hybrid approach to select pre-diversified assets, predict the returns and optimize the portfolio, where the techniques of two-stage clustering, radial basis function neural network (RBFNN) and genetic algorithm were employed. Chen, Zhang, Mehlawat, and Jia (2021) proposed a novel portfolio construction approach based on a hybrid model, termed as IFA-XGBoost, combining eXtreme Gradient Boosting (XGBoost) with an improved firefly algorithm (IFA) for stock prediction and MV model for portfolio selection. Ma, Han, and Wang (2021) combined return prediction in portfolio formation with two machine learning models RF and SVR, and three deep learning models, i.e., LSTM, deep multilayer perceptron (DMLP) and convolutional neural network (CNN). Zhou, Gao, Xiao, Wang, and Liu (2021) designed a stock selection scheme integrating Data Envelopment Analysis (DEA) with multiple data sources, and used Support Vector Machine (SVM) combined with the multi-source data on stocks to predict the stock price movements and combined the obtained stock price movements and the proposed stock selection scheme to construct the portfolio optimization model. Du (2022) established an MV portfolio optimization model using stationary pairs trading portfolios that can be predicted by attention-based LSTM network (A-LSTM). These studies demonstrate that it is necessary and effective to improve and extend the portfolio optimization model by stock forecasting.

This paper proposed a novel two-stage method for well-diversified portfolio construction based on stock return prediction using machine learning. To be specific, the purpose of the first stage is to select diversified stocks with high predicted returns, where the returns are predicted by machine learning methods, i.e. RF, SVR, LSTM, ELM, and BPNN, and the diversification level is measured by Pearson correlation coefficient. In the second stage, the predictive results are incorporated into the modified mean–variance (MMV) model based on open, high, low and close prices to determine the proportion of each asset. Several new characteristics of the proposed method are provided by comparing with existing literature on portfolio selection (see Table 1). In this regard, this paper has three major contributions: (1) stock prediction based on machine learning model is incorporated into the portfolio

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Table 1
Comparison with existing work.

Attribute	Mishra et al. (2016)	Paiva et al. (2019)	Yu et al. (2020)	Yun et al. (2020)	Wang et al. (2020)	Chen et al. (2020)	Chen, Zhang, et al. (2021)	Ma et al. (2021)	Zhou et al. (2021)	Du (2022)	Proposed approach
Portfolio selection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Stock prediction	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Pre-diversification	×	×	×	✓	×	✓	×	×	×	×	✓
Sharpe ratio	×	✓	✓	×	✓	✓	✓	×	✓	✓	✓
Transaction cost	×	✓	✓	✓	✓	×	✓	✓	×	✓	✓
Financial market	Hang-Seng, DAX100, FTSE100, S&P100 and Nikkei225	Ibovespa	ETFs	ETFs	FTSE100	SSE50	SSE50	CSI300	CSI300	CSI300, S& P500	CSI100
Prediction method	HFLANN	SVM	ARIMA	LSTM	LSTM	RBFNN	IFA-XGBoost	RF, SVR, LSTM, CNN, DMLP, ARIMA	SVM	A-LSTM	RF, SVR, LSTM, BPNN, ELM
Portfolio model	PBMV model	MV model	MV, DSR, CVaR, LVaR, MAD, Omega models	GEM model	MV model	MVSK model	MV model	MV, Omega models	MV model	MV model	MMV model
Diversification measure	–	–	–	Pearson correlation coefficient	–	Cluster	–	–	–	–	Pearson correlation coefficient

selection, which not only utilizes the advantages of machine learning in forecasting, but also retains the essence of portfolio model; (2) in order to limit the losses caused by prediction error, this paper proposes a diversified portfolio selection method, that is, buying low-correlation stocks and spreading funds across different assets to minimize the risks caused by insufficient diversification; and (3) we apply the normalized open, high, low, and close prices as input indicators for stock return prediction and variance measure in the MMV model.

The remainder of this paper is organized into six sections. Some machine learning models are presented in Section 2. Section 3 describes the classic MV model and the proposed MMV model. The details of the experiments are introduced in Section 4. Section 5 shows the experimental results. Section 6 outlines the conclusion and possible extensions of future works.

2. Prediction method

2.1. Random forest (RF)

RF developed by Breiman utilizes an ensemble of many decision trees (Breiman, 2001). Generally, if every sample has M features in the dataset, $m = \sqrt{M}$ features are randomly chosen to grow each tree. After constructing decision trees, a significant amount of voting methods are used to determine the final prediction results. Since the RF does not need to assume prior distribution and is easy to deploy, it is suitable for stock prediction problems (Patel, Shah, Thakkar, & Kotecha, 2015).

2.2. Support vector regression (SVR)

SVR is a supervised learning model with the ability of regression (Cortes & Vapnik, 1995). The introduction of kernel makes it possible to handle various types of input data. The goal of the SVR is to find a function that approximates the training instance by minimizing the forecasting error. It can process continuous values and obtain the optimal hyperplane to estimate dependent variables. The model can be expressed by the following formula:

$$\begin{aligned} &min\ \frac{1}{2}\|\omega\|^2 + C\sum_{i=1}^{n}\left(\xi_i + \xi_i{}^*\right) \\ &s.t.\begin{cases} y_i - \omega\cdot x_i - b \le \varepsilon + \xi_i, \\ \omega\cdot x_i + b - y_i \le \varepsilon + \xi_i{}^*, \\ \xi_i, \xi_i{}^* \ge 0, \end{cases} \end{aligned} \tag{1}$$

where ξ, $\xi*$ are slack variables, C and b are cost constant and deviation constant, respectively, and ω represents the vector of the weight coefficient.

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2.3. Long short-term memory network (LSTM)

LSTM is a kind of recurrent neural networks (RNNs), which was proposed by Hochreiter and Schmidhuber (Hochreiter & Schmidhuber, 1997). Compared with other RNNs, it has advantages in retaining information for a long time. The LSTM network consists of input layer, hidden layer, and output layer. Its main feature is the memory cells in the hidden layer. Each memory cell has three gates to maintain and adjust its cell state, namely, forget gate (f_t), input gate (i_t) and output gate (o_t). The formulas for each state and gate are expressed as follows:

$$f_t = sigmoid\left(W_{f,x}x_t + W_{f,h}h_{t-1} + b_f\right), \tag{2}$$

$$i_t = sigmoid\left(W_{i,x}x_t + W_{i,h}h_{t-1} + b_i\right), \tag{3}$$

$$o_t = sigmoid\left(W_{o,x}x_t + W_{o,h}h_{t-1} + b_o\right), \tag{4}$$

$$\tilde{C}_t = \tanh(W_{c,x}x_t + W_{c,h}h_{t-1} + b_c), \tag{5}$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t, \tag{6}$$

$$h_t = o_t * \tanh(C_t), \tag{7}$$

where W_f, W_i, W_o and W_c are weight matrices, x_t and h_t denote the input and hidden state at time t, respectively, b_f, b_i, b_o and b_c are the bias vectors, x_i represents the input variables and C_t is the cell state vector.

2.4. Back propagation neural network (BPNN)

BPNN is a multi-layer feed-forward network based on error back propagation algorithm. It adjusts the weights and thresholds of the neural network using gradient descent method to achieve any small approximation. The BPNN has three or more layers, including input layer, output layer and hidden layer. With the assumption that there are m input neurons, n hidden neurons and s output neurons, the output of all neurons in the hidden (output) layer are calculated as follows:

$$Net_j = \sum_{i=0}^{m} v_{ij}x_i, \tag{8}$$

$$Net_k = \sum_{j=0}^{n} w_{jk}y_j, \tag{9}$$

$$H_j = f(Net_j), \tag{10}$$

$$O_k = f(Net_k), \tag{11}$$

where Net_j (Net_k) represents the activation value of the j_{th} (k_{th}) node, H_j (O_k) is the output of the hidden (output) neuron, and f_v (f_w) is the activation function:

$$f(x) = \frac{1}{1 + \exp(-x)}. \tag{12}$$

2.5. Extreme learning machine (ELM)

ELM proposed by Huang et al. is regarded as a special case of feedforward neuron networks (FNNs) (Huang, Zhu, & Siew, 2006). Instead of completely adjusting the internal parameters like BPNN, ELM randomly assigns input weights and hidden layer biases. The output matrix and the weight of the hidden layer are determined by the inverse of pseudo matrix, which reduces the computational complexity. During the learning process, ELM can skip the iterative adjustment process to adjust the input weights like the traditional neural network. The ELM is formulated as follows:

$$f_{L(x)} = \sum_{i=1}^{L} \beta_i h_i(x) = h(x)\beta, \tag{13}$$

where $\beta = \left[\beta_1, \beta_2, \ldots, \beta_L\right]^T$ represents the output vector and $h(x) = \left[h_1(x), h_2(x), \ldots, h_L(x)\right]$. The sigmoid function is selected as the activation function, so $h(x)$ can be defined as:

$$h_i(x) = sigmoid(\omega_{i0} + \omega_{i1}x_{i1} + \omega_{i2}x_{i2} + \cdots + \omega_{it}x_{it}). \tag{14}$$

3. Modified mean–variance model

This section proposes a modified mean–variance (MMV) model that uses predicted returns as expected returns instead of historical average returns; and uses open, high, low and close (OHLC) prices to calculate the variance rather than only close prices. In the following, the standard MV model is reviewed first, and then the MMV model is proposed.

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In the MV model, the return can be represented as follows:

$$r_p = \sum_{i=1}^{n} w_i r_i, \tag{15}$$

where r_p and r_i are expected returns of the portfolio and asset i, respectively, w_i is the proportion of assets i, and n is the number of assets. The risk is measured by the volatility of returns, and its mathematical expression is:

$$\sigma_p{}^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}, \tag{16}$$

where $\sigma_p{}^2$ represents the variance of the portfolio, w_i (w_j) is the proportion of assets i (j), and σ_{ij} is the covariance between assets i and j.

However, using historical average returns as expected returns has a low-pass filtering effect on the stock market, resulting in inaccurate measurement of expected returns (Freitas, Souza, & de Almeida, 2009). Therefore, we input the indicators computed by OHLC prices into machine learning models to predict stock returns, and then take predicted returns as expected returns in the MMV model:

$$r_e = \sum_{i=1}^{n} w_i \hat{r}_i, \tag{17}$$

where $\hat{r}_i$ and r_e are the predicted return of asset i and the portfolio return, respectively, and n is the number of assets.

Additionally, the volatility of returns is not only reflected in close prices, but also in open, high, low prices (Liu & Wang, 2019). Some studies demonstrated that the return volatility measured by OHLC prices is more effective than that measured only by daily close prices (Arneric, Matkovic, & Soric, 2019; Balter, 2015; Yang & Zhang, 2000). The covariance of the MMV model can be defined as follows:

$$\hat{\sigma}_{ij}^2 = \sigma_{o,ij}^2 + p\sigma_{c,ij}^2 + (1-p)\sigma_{rs,ij}^2, \tag{18}$$

where $\sigma_{o,ij}^2$ is the covariance of o_i and o_j, $\sigma_{c,ij}^2$ is the covariance of c_i and c_j, and $\sigma_{rs,ij}^2$ is defined as follows:

$$\begin{aligned} \sigma_{rs,ij}^2 &= Cov(RS_i, RS_j), \\ RS_i &= h_i(h_i - c_i) + l_i(l_i - c_i). \end{aligned} \tag{19}$$

Specially, o, c, h and l are the normalized open, close, high and low calculated by Eqs. (20)–(23), where O_t is the opening price of the current period (at time t), C_t is the closing price of the current period (at time t), H_t is the current period's high during the trading interval (between [t,1]), and L_t is the current period's low during the trading interval (between [t,1]).

$$o = \ln(\mathrm{O_t}) - \ln(\mathrm{C}_{t-1}), \tag{20}$$

$$c = \ln(\mathrm{C_t}) - \ln(\mathrm{O_t}), \tag{21}$$

$$h = \ln(\mathrm{H_t}) - \ln(\mathrm{O_t}), \tag{22}$$

$$l = \ln(\mathrm{L_t}) - \ln(\mathrm{O_t}), \tag{23}$$

Referring by Yang and Zhang (2000), the p is defined as:

$$p = \frac{0.34}{1.34 + \frac{n+1}{n-1}}. \tag{24}$$

In summary, we propose a mean–variance model based on OHLC prices, which uses the returns predicted by OHLC prices and covariance calculated by OHLC prices to quantify returns and risks of portfolio, respectively. The proposed MMV model can be described by the following formulas:

$$\begin{aligned} &\max \sum_{i=1}^{n} w_i \hat{r}_i \\ &\min \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \hat{\sigma}_{ij} \\ &\text{s.t.} \begin{cases} \sum_{i=1}^{n} w_i = 1, \\ 0 \le w_i \le 1, \forall i = 1, \dots, n. \end{cases} \end{aligned} \tag{25}$$

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Table 2
The ticker symbols of the experimental stocks.

000001	000568	000725	000776	000895	002024
002142	002304	002475	002594	300015	300059
300498	600000	600009	600015	600016	600028
600030	600031	600036	600048	600104	600276
600340	600519	600585	600837	600887	600999
601006	601009	601111	601166	601169	601186
601318	601328	601398	601601	601628	601668
601688	601800	601818	601857	601933	601998

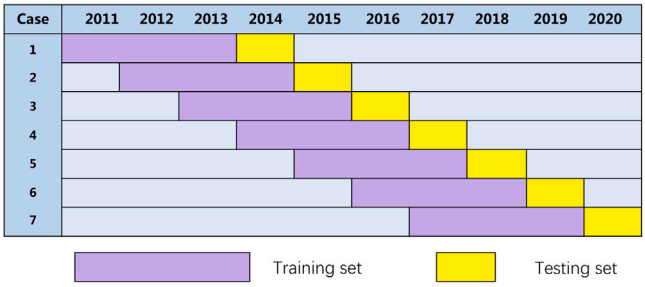

Fig. 1. Rolling window of the experiment. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. The experiment

4.1. Data

This paper use the constituent stocks of China Securities 100 Index as experimental data. The CSI 100 index consists of stocks issued by 100 companies with the largest market capitalization in Shanghai and Shenzhen stock markets. The studying period is daily data from October 8, 2010 to September 30, 2020, and after excluding suspended or unlisted stocks in the selected period, the remaining constituent stocks are 48. Table 2 presents the ticker symbols of these stocks. Due to the continuity of time series data, the experiment is conducted by rolling window. Following Fischer and Krauss (2018b), we consider each "study period" as a training–testing set, including a training set of 3 years and a testing set of 1 year. In each study period, the first 3 years of data is used for rolling window training, and the last 1 year of completely out-of-sample data is tested based on training parameters. We divide the sample data from October 2010 to September 2020 into seven training–testing periods, as shown in Fig. 1. The blue area indicates the entire span of data. The purple area represents the training period, 3 years, and the yellow area is the testing period, 1 year.

4.2. Proposed model: machine learning+MMV

4.2.1. Input indicator

The choice of input indicators is extremely important in time series forecasting task. Since stock returns are affected by many factors, using a single indicator as input cannot achieve good results. Based on previous studies (Paiva et al., 2019; Wang et al., 2020), the indicator combination summarized in Table 3 are widely used as the input indicators. Therefore, we choose these 15 indicators and 4 technical indicators calculated by OHLC prices as the input of machine learning models.

4.2.2. Experimental process

The experimental process has two key stages, namely stock selection (for choosing diversified stocks with high predicted returns) and asset allocation (for spreading funds across selected stocks), as shown in Fig. 2.

Stage 1: stock selection

Input indicators for stock return prediction are calculated by open, high, low and close prices. Then, some machine learning models, including the RF, SVR, LSTM, BPNN and ELM, are applied to predict stock returns. In addition, Pearson correlation coefficient is introduced to measure the diversification level of the portfolio. To construct a profitable and diversified portfolio, we select stocks whose

(A) predicted returns are ranked top k, and

(B) Pearson correlation coefficients with other stocks are less than diversification threshold.

Constraint (A) ensures that the portfolio has profitable returns. Specifically, stocks are ranked in the order of diminishing predicted returns, and the top k ranked stocks are selected. Constraint (B) ensures that the portfolio is sufficiently diversified. For

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Table 3
Input indicators used for stock prediction.

Attribute	Details	Attribute	Details
r_t	$\ln(C_t) - \ln(C_{t-1})$	h_{t-3}	$\ln(H_{t-3}) - \ln(O_{t-3})$
r_{t-1}	$\ln(C_{t-1}) - \ln(C_{t-2})$	l_t	$\ln(L_t) - \ln(O_t)$
r_{t-2}	$\ln(O_{t-2}) - \ln(C_{t-1})$	l_{t-1}	$\ln(L_{t-1}) - \ln(O_{t-1})$
r_{t-3}	$\ln(O_{t-3}) - \ln(C_{t-1})$	l_{t-2}	$\ln(L_{t-2}) - \ln(O_{t-2})$
c_t	$\ln(C_t) - \ln(O_t)$	l_{t-3}	$\ln(L_{t-3}) - \ln(O_{t-3})$
c_{t-1}	$\ln(C_t) - \ln(O_{t-1})$	True range	$\max\{\lvert H_t - L_t\rvert, \lvert C_{t-1} - H_t\rvert, \lvert C_{t-1} - L_t\rvert\}$
o_t	$\ln(O_t) - \ln(C_{t-1})$	Average true range	$MA(TR, N)$
h_t	$\ln(H_t) - \ln(O_t)$	Momentum	$C_t - C_{t-n-1}$
h_{t-1}	$\ln(H_{t-1}) - \ln(O_{t-1})$	Relative strength index	$100 - \frac{100}{1+(\frac{\sum_{i=0}^{N-1} UP_{t-i}}{N})/(\frac{\sum_{i=0}^{N-1} DP_{t-i}}{N})}$, where $UP_t = \max\{0, C_t - C_{t-1}\}$ $DP_t = \max\{0, C_{t-1} - C_t\}$
h_{t-2}	$\ln(H_{t-2}) - \ln(O_{t-2})$		

*O_t, H_t, L_t, C_t are the open, high, low, close prices at time t, MA represents the moving average of the price over N days.

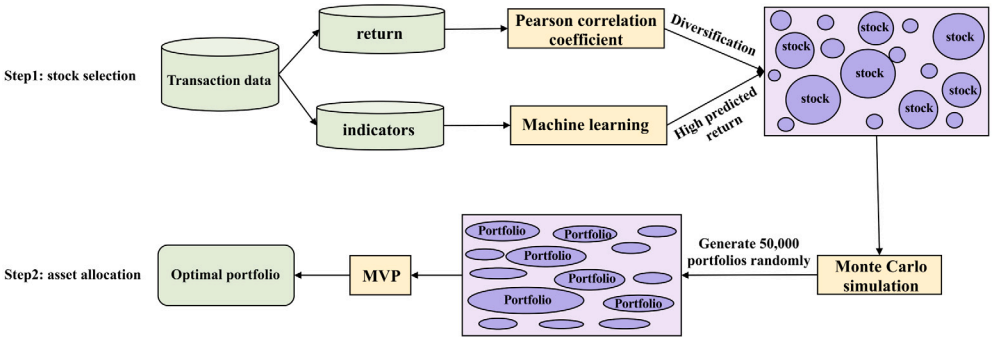

Fig. 2. Overall framework of two-stage portfolio selection method.

Algorithm 1 stock selection.

```
Rank the stocks according to the predicted returns
Initialize the number of the selected stocks n = 0 and stock ranking i = 0
Calculate Pearson correlation coefficients ρ_ij
while n < k do
    i, j = i_max
    for i = i + 1 to i + k − n do
        for j = j + 1 to j + k − n do
            if ρ_ij > θ and i > j then
                Higher-ranked stock is selected and lower-ranked stock is excluded
            else
                Stock i and stock j are selected
            end if
        end for
    end for
    Obtain n
end while
Obtain diversified stocks with high predicted returns
```

example, enforcing diversification threshold $\theta = 0.6$ means that Pearson correlation coefficients between all selected stocks should be less than 0.6. The pseudo code of this process is described as Algorithm 1.

Stage 2: asset allocation

Once high-quality stocks are selected, the next stage is to spread a certain amount of wealth among selected assets. This stage aims to determine the proportion of each asset using the MMV model. Specifically, a set of weights is randomly generated using the Monte Carlo simulation method. Under these weights, mean and variance of portfolio are computed according to the MMV model. Repeat this process 50,000 times, which can cover most possible portfolios with different weights from the statistical perspective. In order to consider both returns and risks, the Sharpe ratio is applied to choose the optimal portfolio, and available funds are allocated to the portfolio with the highest Sharpe ratio.

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Table 4
The considered values of hyperparameters.

Model	Hyperparameter	Value
RF	max_depth	5, 10, 15, 20, 25, 30
	min_samples_split	2, 5, 10, 15, 20, 25, 30
	min_samples_leaf	1, 5, 10, 15, 20, 25, 30
	n_estimators	100, 200, 300, 400, 500
SVR	C	2^0, 2^1, 2^2, 2^3, 2^4, 2^5
	γ	2^{-5}, 2^{-4}, 2^{-3}, 2^{-2}, 2^{-1}, 2^0
	kernel	rbf
LSTM	units	16, 32, 64, 128
	dropout_prob	0.1, 0.2, 0.3, 0.4, 0.5
	batch_size	50, 100, 200
	epochs	100, 200, 300
	optimizer	Adam, SGD, RMSprop
	loss	Mean absolute error, Mean square error
BPNN	units	16, 32, 64, 128
	activation	relu, linear, tanh
	batch_size	50, 100, 200
	epochs	100, 200, 300
	optimizer	Adam, SGD, RMSprop
	loss	Mean absolute error, Mean square error
ELM	hidden nodes	16, 32, 64, 128
	activation	sigmoid

5. Results

5.1. Stock return prediction results

5.1.1. Determination of model hyperparameters

Hyperparameters of machine learning models directly affect its performance. Only when the same level of hyperparameter adjustment procedure is executed, different machine learning methods can be compared fairly. Therefore, the same Grid research method is applied to dynamically find the best hyperparameter combination of different models. Then, machine learning models with the optimal hyperparameters are used to forecast stock returns. Table 4 presents the considered values of hyperparameters.

5.1.2. Evaluation of predicted performance

To evaluate the performance of different machine learning models in stock prediction, Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and Hit Rates (H_R), are chosen, among which, the lower MSE, RMSE and MAE, the higher HR, indicating the more accurate the prediction. The definitions of these metrics are as follows:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(r_i - \hat{r}_i)^2, \tag{26}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(r_i - \hat{r}_i)^2}, \tag{27}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|r_i - \hat{r}_i|, \tag{28}$$

$$\mathrm{H_R} = \frac{1}{n}\sum_{i=1}^{n}a_i, \tag{29}$$

where r_i represents the actual returns, $\hat{r}_i$ indicates the predicted returns, n is the total number of forecasting periods, $a_i = 1$ if $r_i\hat{r}_i > 0$ and $a_i = 0$ otherwise.

Table 5 summarizes the prediction results of each model from 2014 to 2020. For the MSE, RMSE and MAE, the RF performs better than other models in most years, but there are some exceptions. For example, the MAE of the RF model is higher than the LSTM model in 2016, and the MSE, MAE and RMSE of the RF model are larger than the LSTM model in 2017. For the H_R metric, although the performance of RF is not always the best, it is still better than most other models. For example, in 2017, SVR has the greatest H_R of 49.14%, followed by the RF with 48.93%. Similarly, in 2020, RF follows the greatest H_R of LSTM with 48.42%.

The mean MSE, RMSE, MAE, and H_R of different machine learning models are presented in Table 6. It is obvious that the RF model has better prediction performance than others and the SVR model, as well as the LSTM model, is second only to the RF model but superior to the ELM model and BPNN model. For example, the average MSE value of the RF, SVR, LSTM, ELM, BPNN are 5.26E−04, 5.37E−04, 5.46E−04, 5.46E−04, 5.70E−04, respectively. The RF compared with the other models is reduced by 2.1%, 3.8%, 3.8%, and 8.4% in terms of MSE. Similarly, the reduction percentage of the RMSE are 0.93%, 0.93%, 1.87% and

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Table 5
Annual forecast performance of different models.

	MSE	RMSE	MAE	H_R	MSE	RMSE	MAE	H_R
	2014				2015			
RF	3.39E−04	1.76E−02	1.28E−02	47.35%	1.35E−03	3.62E−02	2.64E−02	49.74%
SVR	3.42E−04	1.77E−02	1.29E−02	47.03%	1.38E−03	3.66E−02	2.67E−02	49.67%
LSTM	3.40E−04	1.76E−02	1.27E−02	48.73%	1.45E−03	3.75E−02	2.72E−02	48.43%
ELM	3.56E−04	1.81E−02	1.32E−02	48.35%	1.37E−03	3.66E−02	2.67E−02	49.23%
BPNN	3.58E−04	1.82E−02	1.34E−02	47.84%	1.51E−03	3.83E−02	2.81E−02	48.04%
	2016				2017			
RF	4.64E−04	2.09E−02	1.44E−02	47.81%	2.25E−04	1.45E−02	1.02E−02	48.93%
SVR	4.77E−04	2.12E−02	1.46E−02	48.13%	2.30E−04	1.47E−02	1.04E−02	49.14%
LSTM	4.73E−04	2.10E−02	1.42E−02	47.85%	2.24E−04	1.44E−02	1.01E−02	48.39%
ELM	4.90E−04	2.14E−02	1.50E−02	47.77%	2.40E−04	1.50E−02	1.07E−02	48.09%
BPNN	4.94E−04	2.15E−02	1.50E−02	46.82%	2.40E−04	1.50E−02	1.09E−02	47.32%
	2018				2019			
RF	4.45E−04	2.04E−02	1.51E−02	47.19%	4.22E−04	1.98E−02	1.44E−02	48.63%
SVR	4.55E−04	2.06E−02	1.53E−02	47.13%	4.29E−04	2.00E−02	1.47E−02	48.39%
LSTM	4.54E−04	2.05E−02	1.51E−02	48.01%	4.30E−04	2.00E−02	1.44E−02	49.18%
ELM	4.69E−04	2.09E−02	1.56E−02	48.28%	4.43E−04	2.03E−02	1.49E−02	48.16%
BPNN	4.71E−04	2.10E−02	1.56E−02	47.03%	4.52E−04	2.06E−02	1.51E−02	47.08%
	2020							
RF	4.37E−04	2.00E−02	1.42E−02	48.42%				
SVR	4.48E−04	2.03E−02	1.45E−02	47.93%				
LSTM	4.49E−04	2.03E−02	1.42E−02	49.44%				
ELM	4.55E−04	2.04E−02	1.46E−02	47.86%				
BPNN	4.63E−04	2.07E−02	1.50E−02	47.34%				

Table 6
Average forecast performance of different models.

	MSE	RMSE	MAE	H_R
RF	5.26E−04	2.14E−02	1.53E−02	48.29%
SVR	5.37E−04	2.16E−02	1.56E−02	48.19%
LSTM	5.46E−04	2.16E−02	1.54E−02	48.57%
ELM	5.46E−04	2.18E−02	1.58E−02	48.25%
BPNN	5.70E−04	2.22E−02	1.62E−02	47.35%

3.74%, respectively, as well as the reduction percentage of the MAE are 1.96%, 0.65%, 3.27%, 5.88%, respectively. For directional evaluation index H_R, the RF model is increased by 0.21%, 0.08%, and 1.95% compared with the SVR, ELM, and BPNN. The performance of LSTM is only better than RF in terms of H_R. These results may attribute to the simple structure of our neural network models (only use a 2-hidden-layer module). Note that the performance of the neural network may be better than the RF model if the complex structure is used. Our results are consistent with Patel et al. (2015) and Ma et al. (2021), who also found that the RF model is superior to the SVR model and the LSTM model.

5.2. Portfolio selection results

In this section, we select three machine learning models, RF, SVR, and LSTM, that perform better in the previous section for further comparison. Moreover, since it is hard for individual investors to manage too many different stocks, this paper attempts to construct a portfolio with seven stocks ($k = 7$) according to Chen, Zhang, et al. (2021). In the following, parameters in the portfolio construction process are determined first, and then the results of portfolio selection are presented.

5.2.1. Parameter settings

(1) Diversification threshold

In this section, we explore the setting of diversification thresholds that enable a portfolio to achieve higher returns and lower risks. In order to quantify the relationship between stocks, this paper uses the Pearson correlation coefficient:

$$\rho_{ij} = \frac{\sum_t \left[(r_i(t) - \langle r_i \rangle)(r_j(t) - \langle r_j \rangle)\right]}{\sqrt{\sum_t (r_i(t) - \langle r_i \rangle)^2}\sqrt{\sum_t (r_j(t) - \langle r_j \rangle)^2}}, \tag{30}$$

where $r_i(t)$ and $r_j(t)$ are the returns of stock i and j on t day, respectively, and $\langle \cdots \rangle$ represents the mean value.

This paper set the diversification threshold θ as 0.3, 0.4, 0.5, 0.6, 0.7 and 1 (without considering diversification). Furthermore, in order to avoid the randomness brought by solving the MMV model and MV model, the proportion of each asset is set as equal

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Table 7
Comparison of different diversification threshold.

		RF					
		1	0.7	0.6	0.5	0.4	0.3
A	2014	0.0684	0.1614	0.1397	0.0706	0.0557	0.0186
	2015	1.1886	1.0282	1.2344	1.2919	1.0551	1.0553
	2016	0.1661	0.1843	0.1470	0.1032	0.1402	0.1846
	2017	0.6028	0.5587	0.6364	0.5070	0.4540	0.3313
	2018	0.1094	0.1002	0.1531	0.1442	0.1941	0.0902
	2019	0.2894	0.2674	0.2735	0.1913	0.1808	0.1936
	2020	0.2621	0.2412	0.2409	0.1178	0.1717	0.1686
	Avg.	0.3838	0.3630	0.4036	0.3466	0.3217	0.2917
B	2014	0.2014	0.2026	0.1993	0.1955	0.1912	0.1949
	2015	0.4632	0.4601	0.4543	0.4494	0.4488	0.4548
	2016	0.2936	0.2900	0.2888	0.2903	0.2851	0.2830
	2017	0.1444	0.1412	0.1447	0.1393	0.1336	0.1393
	2018	0.2465	0.2446	0.2490	0.2421	0.2392	0.2309
	2019	0.2542	0.2535	0.2528	0.2582	0.2639	0.2574
	2020	0.2719	0.2657	0.2654	0.2610	0.2562	0.2445
	Avg.	0.2679	0.2654	0.2649	0.2623	0.2597	0.2578
C	2014	0.1909	0.6486	0.5506	0.2077	0.1346	−0.0583
	2015	2.5014	2.1694	2.6508	2.8078	2.2842	2.2542
	2016	0.4637	0.5322	0.4051	0.2523	0.3867	0.5462
	2017	3.9652	3.7449	4.1894	3.4256	3.1739	2.1630
	2018	0.3220	0.2869	0.4944	0.4716	0.6860	0.2606
	2019	1.0205	0.9367	0.9631	0.6245	0.5715	0.6356
	2020	0.8534	0.7947	0.7947	0.3366	0.5531	0.5669
	Avg.	1.3310	1.3019	1.4355	1.1609	1.1129	0.9097

Table 8
Comparison of different diversification threshold.

		SVR					
		1	0.7	0.6	0.5	0.4	0.3
A	2014	0.2305	0.2267	0.2368	0.2243	0.1347	0.2594
	2015	1.0896	1.1696	1.2051	1.1823	1.1677	1.3233
	2016	0.2008	0.2087	0.1953	0.1324	0.1133	0.1075
	2017	0.1555	0.1921	0.2694	0.2916	0.2849	0.2927
	2018	0.1100	0.1440	0.2005	0.2658	0.1560	0.2447
	2019	0.3467	0.3311	0.3217	0.3141	0.2309	0.1985
	2020	0.2442	0.2434	0.2341	0.2048	0.2223	0.3084
	Avg.	0.3396	0.3594	0.3804	0.3736	0.3300	0.3907
B	2014	0.1880	0.1839	0.1844	0.1930	0.1899	0.1751
	2015	0.4580	0.4540	0.4459	0.4441	0.4361	0.4351
	2016	0.2931	0.2857	0.2840	0.2824	0.2718	0.2688
	2017	0.1223	0.1217	0.1218	0.1244	0.1284	0.1240
	2018	0.2277	0.2267	0.2296	0.2239	0.2145	0.2179
	2019	0.2370	0.2359	0.2351	0.2370	0.2319	0.2264
	2020	0.2405	0.2382	0.2391	0.2420	0.2419	0.2254
	Avg.	0.2524	0.2494	0.2486	0.2495	0.2449	0.2389
C	2014	1.0666	1.0699	1.1210	1.0067	0.5514	1.3099
	2015	2.3137	2.5102	2.6355	2.5948	2.6086	2.9727
	2016	0.5827	0.6256	0.5819	0.3627	0.3066	0.2885
	2017	1.0265	1.3319	1.9653	2.1038	1.9848	2.1190
	2018	0.3516	0.5029	0.7427	1.0530	0.5876	0.9853
	2019	1.3358	1.2764	1.2408	1.1991	0.8664	0.7445
	2020	0.8905	0.8957	0.8534	0.7224	0.7948	1.2353
	Avg.	1.0811	1.1732	1.3058	1.2918	1.1000	1.3793

weight (1/N) here. Tables 7 to 9 presents the return (Panel A), risk (Panel B) and Sharpe ratio (Panel C) of portfolios under different thresholds, and Figs. 3 to 5 visualize these results.

As can be seen from the results, diversified portfolios tend to have higher returns and lower risks. In particular, when the threshold is 0.6, mean returns and Sharpe ratios of RF and SVR models are greater than those without considering diversification ($\theta = 1$), and standard deviations of all models are less than those with the threshold $\theta = 1$. To sum up, setting the diversification threshold to 0.6 gives a better trade-off among (i) stock diversification, (ii) portfolio returns, and (iii) portfolio risks. Consequently, we focus subsequent analysis on the diversification threshold of 0.6.

(2) Days for computing the covariance matrix

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Table 9
Comparison of different diversification threshold.

		LSTM					
		1	0.7	0.6	0.5	0.4	0.3
A	2014	0.2631	0.2340	0.2429	0.2260	0.1061	0.0633
	2015	0.8354	0.7670	0.6830	0.7034	0.6231	0.4757
	2016	0.2299	0.1611	0.2036	0.2081	0.1443	0.0812
	2017	0.4561	0.4704	0.4164	0.4892	0.4378	0.4782
	2018	0.0647	0.0600	0.0085	0.0190	0.0970	0.1978
	2019	0.3935	0.4047	0.4210	0.4188	0.4213	0.3310
	2020	0.3418	0.3297	0.3478	0.3809	0.3749	0.3022
	Avg.	0.3692	0.3467	0.3319	0.3493	0.3149	0.2756
B	2014	0.2025	0.1984	0.2016	0.2025	0.2074	0.1914
	2015	0.4337	0.4361	0.4345	0.4378	0.4365	0.4300
	2016	0.2687	0.2663	0.2680	0.2666	0.2632	0.2526
	2017	0.1349	0.1314	0.1288	0.1312	0.1286	0.1316
	2018	0.2531	0.2524	0.2478	0.2475	0.2417	0.2430
	2019	0.2488	0.2487	0.2473	0.2520	0.2520	0.2467
	2020	0.2608	0.2600	0.2602	0.2630	0.2599	0.2535
	Avg.	0.2575	0.2562	0.2554	0.2572	0.2556	0.2498
C	2014	1.1514	1.0278	1.0564	0.9681	0.3666	0.1737
	2015 1.8571	1.6899	1.5030	1.5382	1.3588	1.0365	
	2016	0.7440	0.4925	0.6478	0.6679	0.4344	0.2029
	2017	3.1592	3.3520	2.9994	3.5009	3.1700	3.4047
	2018	0.1370	0.1189	−0.0867	−0.0445	0.2773	0.6907
	2019	1.4610	1.5065	1.5811	1.5428	1.5528	1.2198
	2020	1.1953	1.1527	1.2216	1.3345	1.3273	1.0740
	Avg.	1.3864	1.3343	1.2746	1.3583	1.2125	1.1146

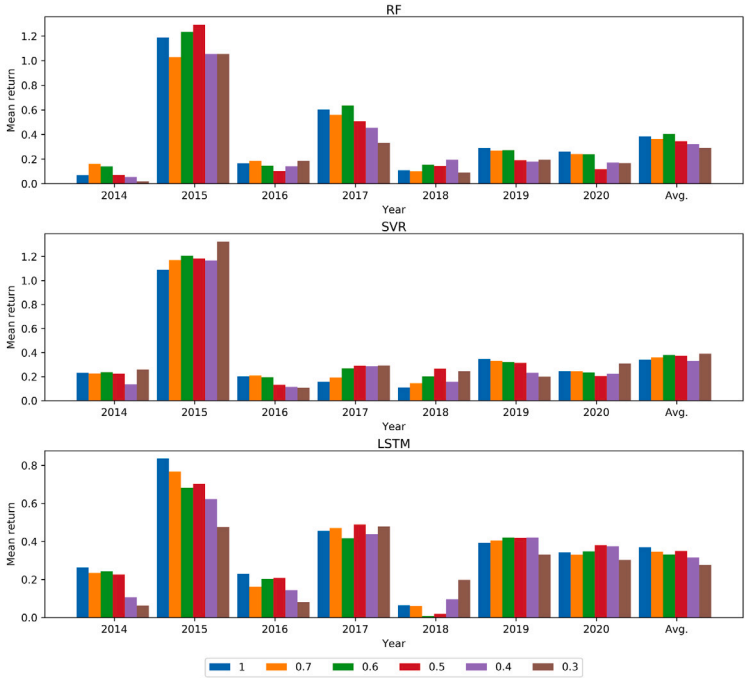

Fig. 3. Annualized return for portfolios under different diversification thresholds.

To obtain the optimal allocation of the MV and the MMV models, days for calculating the covariance matrix are tested. Fig. 6 summarizes the proportion of each model's return in 50, 75, and 100 days. As for the MV model, the covariance matrix calculated with 75 days in the RF achieves the highest return, 50 days in the SVR, and 75 days in the LSTM. Regarding the MMV model, the covariance matrix calculated with 50 days in the RF realizes the highest return, 75 days in the SVR, and 50 days in the LSTM. Therefore, we use 75 days to calculate the covariance matrix of the MV and 50 days to calculate the covariance matrix of the MMV in the follow-up study.

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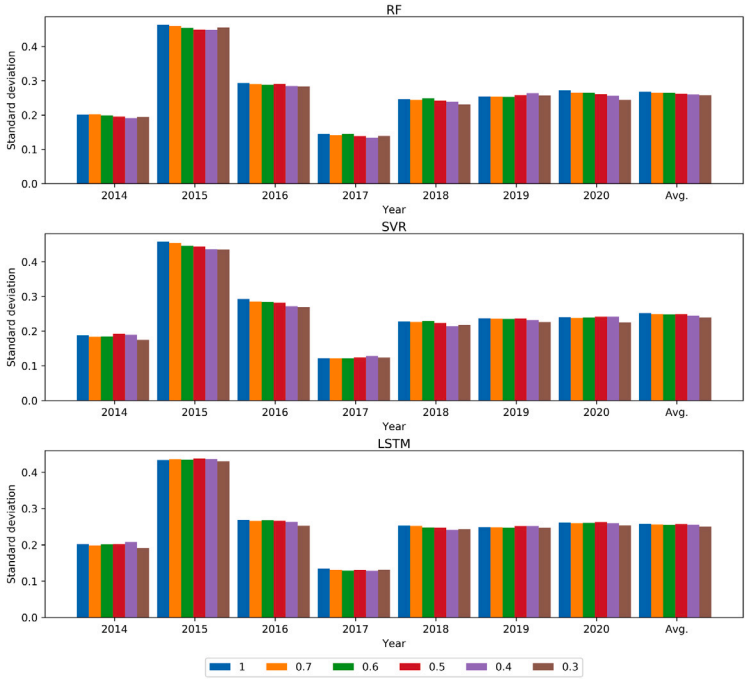

Fig. 4. Risk for portfolios under different diversification thresholds.

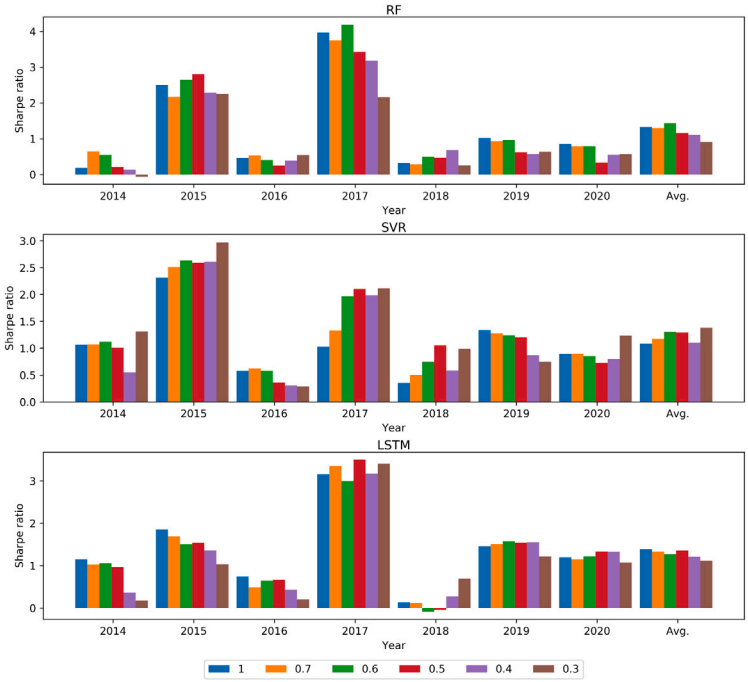

Fig. 5. Sharpe ratio for portfolios under different diversification thresholds.

5.2.2. Model effectiveness

In this section, we discuss the performance of different models before and after considering transaction costs to further compare their profitability in the real stock market. Referring to several empirical studies (Almahdi & Yang, 2017; Paiva et al., 2019), this paper sets transaction costs as 0.05 ‰ and 0.1 ‰. Besides, the MV model and equal weight (1/N) portfolio are widely used in existing studies. Thus, we introduce several similar models and market index to evaluate the effectiveness of the proposed machine

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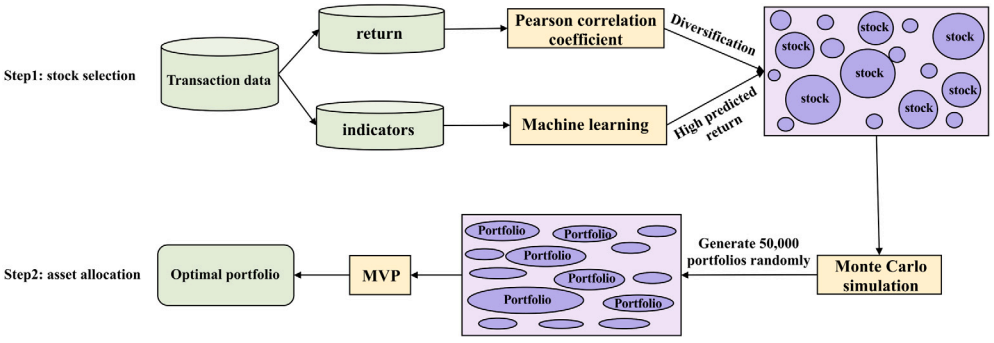

Fig. 6. Days for computing the covariance matrix.

Table 10
Daily return–risk characteristics without transaction cost.

Model	Mean return	Standard dev.	Downside dev.	1-percent VaR	5-percent VaR	1-percent CVaR	5-percent CVaR
RF+MMV	0.0016	0.0179	0.0124	0.0400	0.0278	0.0460	0.0353
RF+MV	0.0014	0.0192	0.0134	0.0432	0.0302	0.0497	0.0382
RF+1/N	0.0012	0.0176	0.0125	0.0397	0.0277	0.0457	0.0351
SVR+MMV	0.0014	0.0171	0.0120	0.0384	0.0267	0.0442	0.0339
SVR+MV	0.0013	0.0183	0.0128	0.0413	0.0288	0.0475	0.0365
SVR+1/N	0.0012	0.0167	0.0119	0.0377	0.0263	0.0434	0.0333
LSTM+MMV	0.0014	0.0173	0.0126	0.0388	0.0270	0.0446	0.0342
LSTM+MV	0.0013	0.0181	0.0131	0.0408	0.0285	0.0470	0.0361
LSTM+1/N	0.0011	0.0170	0.0124	0.0383	0.0268	0.0441	0.0339
CSI100 Index	0.0005	0.0148	0.0108	0.0340	0.0239	0.0390	0.0301

learning+MMV model. Tables 10 to 15 summarizes the performance of each model, where Tables 10, 12 and 14 describe the daily return–risk characteristics, and Tables 11, 13 and 15 describe the annualized return–risk measurement.

Daily return–risk characteristics

As can be seen that the RF+MMV has the highest daily mean return, with 0.0016, before taking into account transaction cost. The second best model is the SVR+MMV, with the average return of 0.0014, and the RF+MV ranked third with the average return of 0.0014 before transaction cost. We find that the mean returns of portfolios constructed by the MMV are greater than those by the MV and the 1/N. This finding holds even after accounting for transaction costs. Furthermore, several metrics are used to assess the portfolio risks, i.e. standard deviation, downside deviation, Value at Risk (VaR), and Conditional Value at Risk (CVaR). Before transaction cost, the riskiest strategy stems from the RF+MV, 0.0192 for standard deviation, 0.0134 for downside deviation, 0.0432 for 1-percent VaR, 0.0302 for 5-percent VaR, 0.0497 for 1-percent CVaR and 0.0382 for 5-percent CVaR, and the least risky strategy comes from the CSI 100 index, as it contains more stocks to diversify risks. After accounting for transaction costs, as shown in Tables 12 to 15, the risks of the MMV-based portfolios are lower than the MV-based portfolios and higher than the 1/N-based portfolios using the same prediction method.

Annualized risk–return metrics

Following Tables 11, 13 and 15, it is obvious that: (1) when transaction costs are ignored, the RF+MMV achieved the best place with annualized mean return of 0.4939, Sharpe ratio of 1.6353, Sortino ratio of 2.3516 and maximum drawdown of 0.2999. Nevertheless, the CSI 100 index has the lowest risk among these models, 0.2354 for standard deviation and 0.1722 for downside deviation; (2) when transaction costs are considered, the RF+MMV has the higher annualized mean return, Sharpe ratio, Sortino ratio and maximum drawdown than other models and the CSI 100 index.

5.2.3. Visualization on financial performance

To evaluate the performance of different models more intuitively, we visualize the portfolio characteristics in various forms. Fig. 7 shows the accumulative returns of nine models and the CSI 100 index without considering transaction costs. It can be seen that the RF+MMV has the highest accumulative return among all the portfolio models, with 10.558. For example, compared with the RF+MV and RF+1/N, the accumulative returns are increased by 3.839 and 5.319, respectively, demonstrating the effectiveness of the MMV model; and compared with the SVR+MMV and LSTM+MMV, the accumulative returns are increased by 3.238 and 3.533, respectively, indicating that the more accuracy the prediction, the better the performance of the portfolio. In addition, as shown in Fig. 8, all prediction-based models, with transaction costs of 0.5 ‰, are superior to the CSI 100 index. After considering the transaction cost of 1 ‰, as shown in Fig. 9, only the RF+MMV outperforms the CSI 100 index.

Moreover, inspired by the studies (Paiva et al., 2019; Wang et al., 2020), we demonstrate the performance of each model over time, i.e. October 2014 to September 2020, for further comparisons. From Fig. 10, of the seven surveyed year, accumulative returns

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Table 11
Annualized risk–return metrics without transaction cost.

Model	Mean return	Standard dev.	Downside dev.	Sharpe ratio	Sortino ratio	Max. drawdown
RF+MMV	0.4939	0.2837	0.1973	1.6353	2.3516	0.2999
RF+MV	0.4161	0.3044	0.2135	1.2685	1.8089	0.3947
RF+1/N	0.3625	0.2794	0.1987	1.1900	1.6731	0.3681
SVR+MMV	0.4183	0.2713	0.1899	1.4313	2.0455	0.3452
SVR+MV	0.3886	0.2909	0.2037	1.2327	1.7605	0.3393
SVR+1/N	0.3485	0.2654	0.1886	1.2001	1.6891	0.3798
LSTM+MMV	0.4121	0.2741	0.1994	1.3942	1.9162	0.3898
LSTM+MV	0.3817	0.2875	0.2078	1.2231	1.6925	0.4233
LSTM+1/N	0.3175	0.2691	0.1972	1.0684	1.4581	0.4127
CSI100 Index	0.1411	0.2354	0.1722	0.4718	0.6450	0.4414

Table 12
Daily return–risk characteristics with transaction cost 0.5 ‰.

Model	Mean return	Standard dev.	Downside dev.	1-percent VaR	5-percent VaR	1-percent CVaR	5-percent CVaR
RF+MMV	0.0011	0.0179	0.0124	0.0405	0.0283	0.0465	0.0358
RF+MV	0.0009	0.0192	0.0134	0.0437	0.0307	0.0502	0.0387
RF+1/N	0.0007	0.0176	0.0125	0.0402	0.0282	0.0462	0.0356
SVR+MMV	0.0009	0.0171	0.012	0.0389	0.0272	0.0447	0.0344
SVR+MV	0.0008	0.0183	0.0128	0.0418	0.0293	0.0480	0.0370
SVR+1/N	0.0007	0.0167	0.0119	0.0382	0.0275	0.0439	0.0338
LSTM+MMV	0.0009	0.0173	0.0126	0.0393	0.029	0.0451	0.0347
LSTM+MV	0.0008	0.0181	0.0131	0.0413	0.0273	0.0475	0.0366
LSTM+1/N	0.0006	0.0170	0.0124	0.0388	0.0239	0.0446	0.0344
CSI100 Index	0.0005	0.0148	0.0108	0.0340	0.0239	0.0390	0.0301

Table 13
Annualized risk–return metrics with transaction cost 0.5 ‰.

Model	Mean return	Standard dev.	Downside dev.	Sharpe ratio	Sortino ratio	Max. drawdown
RF+MMV	0.3173	0.2837	0.1973	1.0127	1.4563	0.3177
RF+MV	0.2487	0.3044	0.2135	0.7183	1.0243	0.4139
RF+1/N	0.2013	0.2794	0.1987	0.6132	0.8621	0.3881
SVR+MMV	0.2506	0.2713	0.1899	0.8131	1.1620	0.3622
SVR+MV	0.2244	0.2909	0.2037	0.6681	0.9542	0.3731
SVR+1/N	0.189	0.2654	0.1886	0.5990	0.8431	0.3959
LSTM+MMV	0.2451	0.2741	0.1994	0.7848	1.0787	0.4050
LSTM+MV	0.2182	0.2875	0.2078	0.6548	0.906	0.4535
LSTM+1/N	0.1616	0.2691	0.1972	0.4892	0.6677	0.4400
CSI100 Index	0.1411	0.2354	0.1722	0.4718	0.6450	0.4414

Table 14
Daily return–risk characteristics with transaction cost 1 ‰.

Model	Mean return	Standard dev.	Downside dev.	1-percent VaR	5-percent VaR	1-percent CVaR	5-percent CVaR
RF+MMV	0.0006	0.0179	0.0124	0.0410	0.0288	0.0470	0.0363
RF+MV	0.0004	0.0192	0.0134	0.0442	0.0312	0.0507	0.0392
RF+1/N	0.0002	0.0176	0.0125	0.0407	0.0287	0.0467	0.0361
SVR+MMV	0.0004	0.0171	0.0120	0.0394	0.0277	0.0452	0.0349
SVR+MV	0.0003	0.0183	0.0128	0.0423	0.0298	0.0485	0.0375
SVR+1/N	0.0002	0.0167	0.0119	0.0387	0.0273	0.0444	0.0343
LSTM+MMV	0.0004	0.0173	0.0126	0.0398	0.0280	0.0456	0.0352
LSTM+MV	0.0003	0.0181	0.0131	0.0418	0.0295	0.0480	0.0371
LSTM+1/N	0.0001	0.0170	0.0124	0.0393	0.0278	0.0451	0.0349
CSI100 Index	0.0005	0.0148	0.0108	0.0340	0.0239	0.0390	0.0301

and Sharpe ratios of the prediction-based models are superior to the CSI 100 index in the corresponding periods, with the RF+MMV in particular. Moreover, portfolios constructed by the RF achieve exceptionally high returns in 2015 and 2017, due to the key advantage of RF's robustness to noise and outliers—which occurs at volatile times. As shown in Figs. 11 and 12, after transaction costs of 0.5 ‰ and 1 ‰, we can see that: (1) From the perspective of different forecasting methods, portfolios predicted by the RF have higher accumulative returns than other models in the seven periods. (2) From the perspective of different portfolio optimization models, portfolios constructed by the MMV have higher Sharpe ratio than those by the MV and 1/N in six out of seven periods. Overall, the RF+MMV model behaves better than other models and CSI 100 index.

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Table 15
Annualized risk–return metrics with transaction cost 1 ‰.

Model	Mean return	Standard dev.	Downside dev.	Sharpe ratio	Sortino ratio	Max. drawdown
RF+MMV	0.1615	0.2837	0.1973	0.4634	0.6664	0.3478
RF+MV	0.1009	0.3044	0.2135	0.2330	0.3322	0.4325
RF+1/N	0.0592	0.2794	0.1987	0.1044	0.1468	0.4860
SVR+MMV	0.1026	0.2713	0.1899	0.2677	0.3826	0.3867
SVR+MV	0.0795	0.2909	0.2037	0.1701	0.2430	0.4508
SVR+1/N	0.0483	0.2654	0.1886	0.0689	0.0969	0.5399
LSTM+MMV	0.0978	0.2741	0.1994	0.2473	0.3400	0.4305
LSTM+MV	0.0741	0.2875	0.2078	0.1534	0.2122	0.5144
LSTM+1/N	0.0242	0.2691	0.1972	−0.0216	−0.0295	0.5836
CSI100 Index	0.1411	0.2354	0.1722	0.4718	0.6450	0.4414

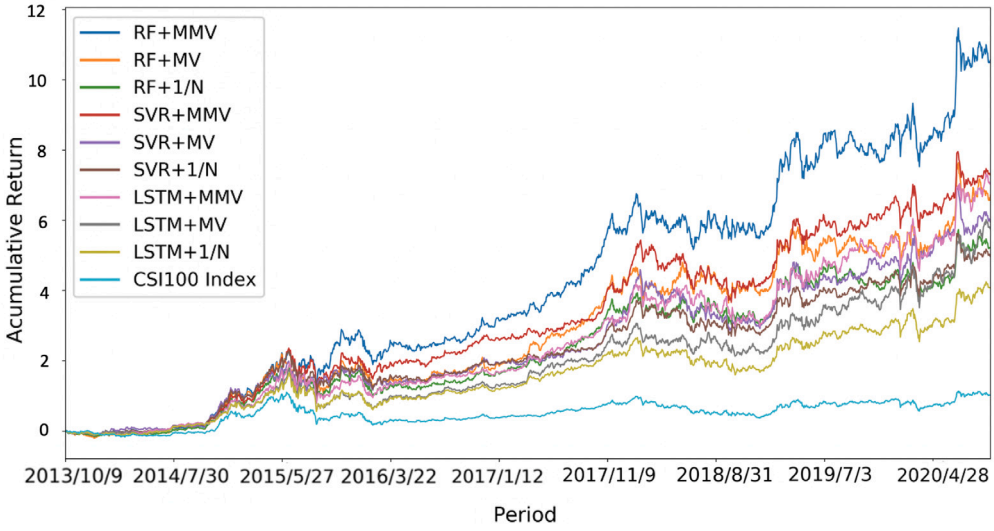

Fig. 7. Accumulative returns without transaction cost.

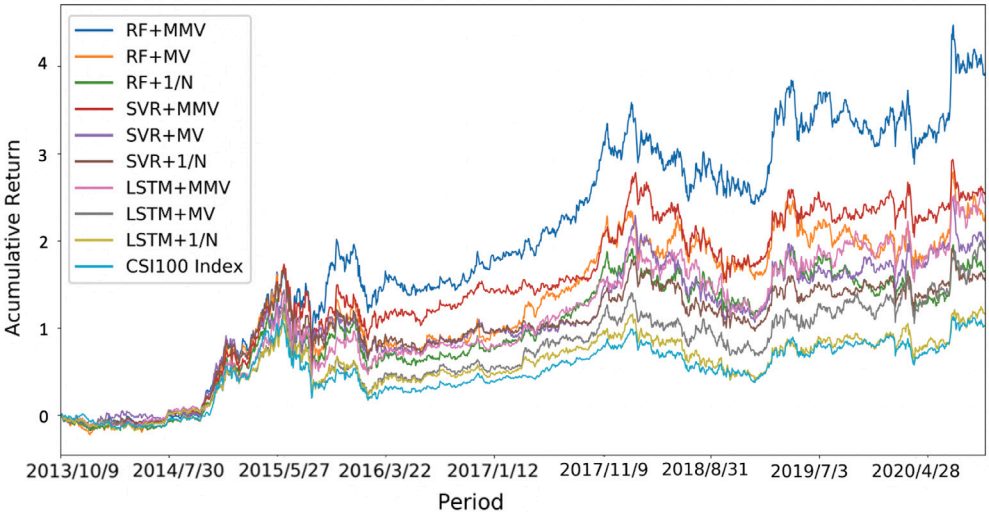

Fig. 8. Accumulative returns with transaction cost (0.5 ‰).

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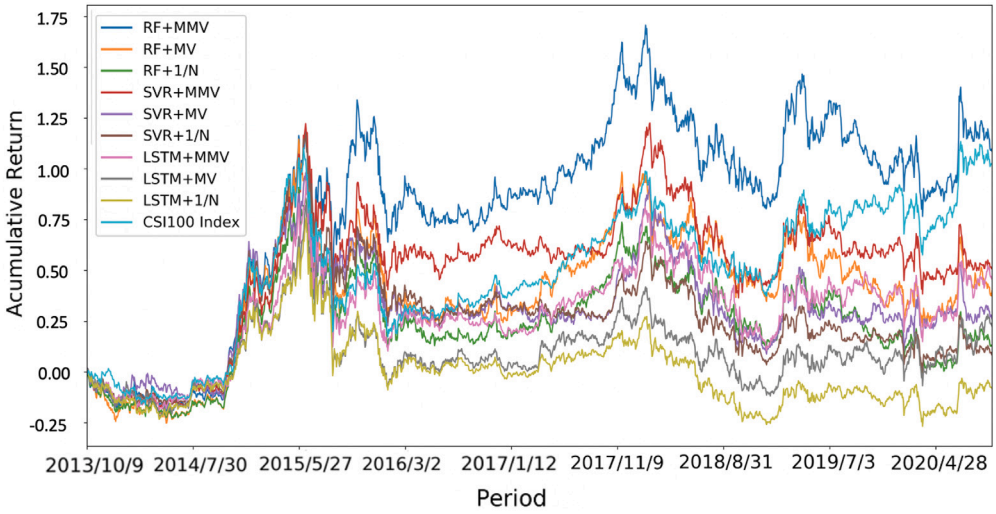

Fig. 9. Accumulative returns with transaction cost (1 ‰).

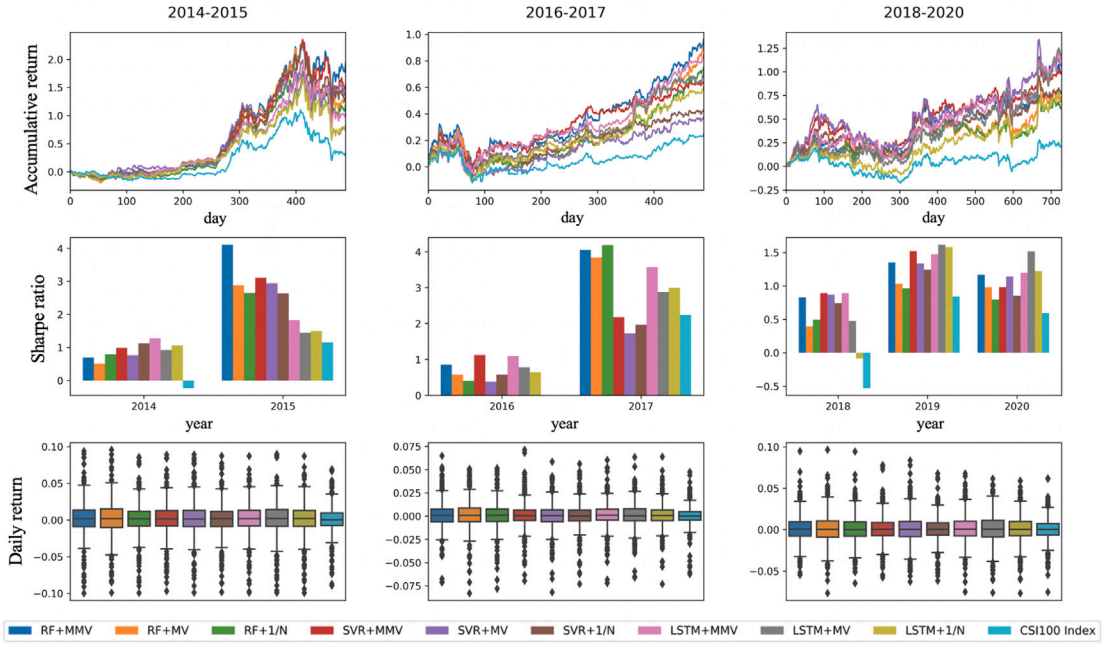

Fig. 10. Contrast of different models from 2014 to 2020 without transaction cost.

6. Discussion and conclusions

This paper proposes a two-stage well-diversified portfolio construction method with stock return prediction. To be specific, the purpose of the first stage is to select diversified stocks with high predicted returns, where the returns are predicted by machine learning methods, i.e. RF, SVR, LSTM, ELM and BPNN, and the diversification level is measured by Pearson correlation coefficient. Then, available funds are allocated to the selected stocks, and the weights are determined by MMV, MV and 1/N models. From the experimental results, we have several important findings:

First of all, four error metrics, i.e. MSE, RMSE, MAE and H_R, are used to evaluate the predictive ability of the RF, SVR, LSTM, ELM and BPNN. Empirical results demonstrate that the RF model is better than other models in predicting stock returns.

Second, as we all know, the prediction error is inevitable, which may bring losses to investors. To limit the losses, an effective strategy is diversification, which involves buying low-correlation stocks and spreading the funds across different assets. In addition,

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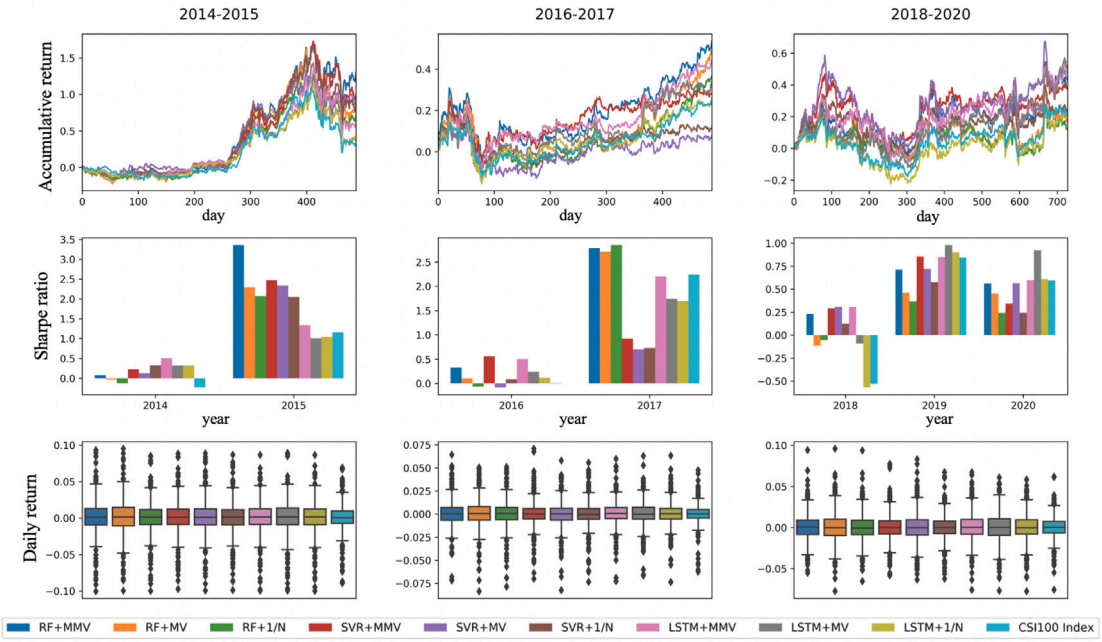

Fig. 11. Contrast of different models from 2014 to 2020 with transaction cost 0.5 ‰.

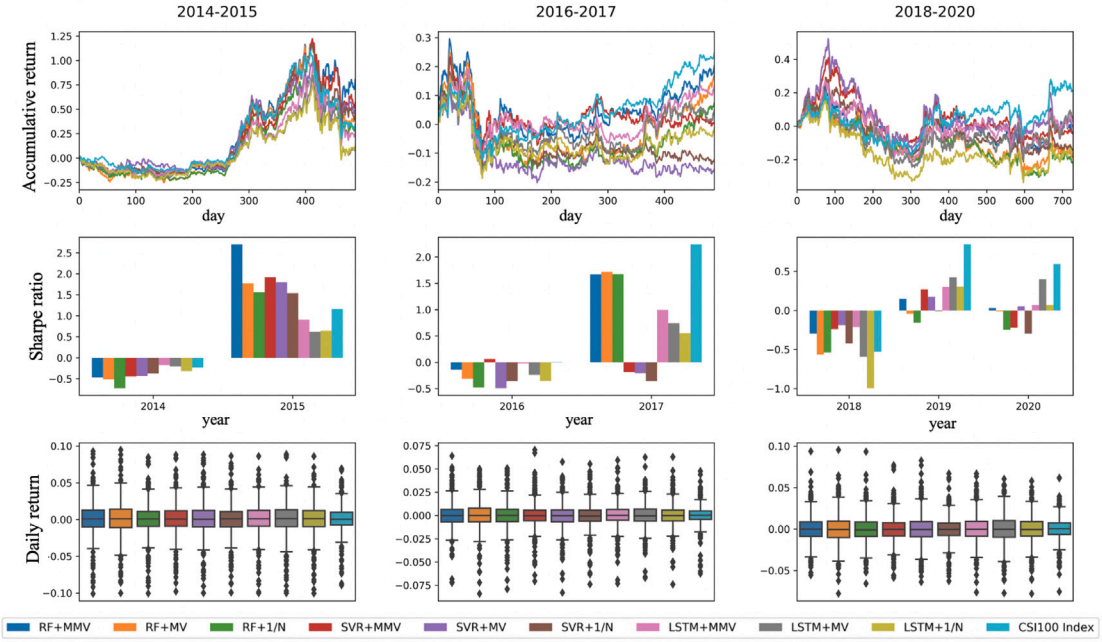

Fig. 12. Contrast of different models from 2014 to 2020 with transaction cost 1 ‰.

we find that the diversification threshold of 0.6 can make the trade-off between maximizing returns and minimizing risks caused by insufficient dispersion.

Third, without transaction costs, machine learning-based portfolio models are significantly better than the CSI 100 index in terms of return and return–risk metrics. Moreover, portfolios based on the MMV model outperform those based on the MV model and 1/N model. After taking transaction costs into account, the RF+MMV model is superior to other models with better results.

These findings suggest that the proposed models, in particular RF+MMV, can be an effective tool for constructing profitable portfolios. Even after deducting the transaction costs, it can still make a substantial return.

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CRediT authorship contribution statement

Wei Chen: Conceptualization, Writing – review and editing, Supervision, Funding acquisition. **Haoyu Zhang:** Conceptualization, Methodology, Software, Writing – original draft. **Lifen Jia:** Methodology, Writing – review and editing.

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